

Lab 2: Projection and Coordinate Systems

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1. Deliverables

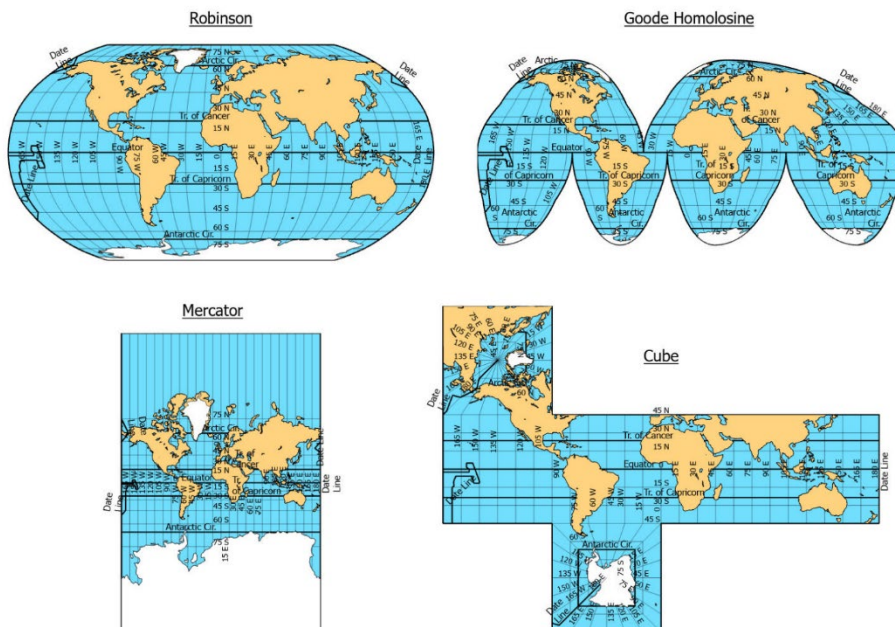
A) The measurements of the area of Philadelphia:

CRS	Area (Acres)
SP Pennsylvania South	91,347.93
NAD 1983 UTM Zone 18N	91,277.12
US National Atlas Equal Area	91,252.59
NAD 1983 Contiguous USA Albers	91,349.78
WGS 1984 World Mercator	155,271.80

- SP Pennsylvania South is accurate because it's a local mapping in PA's southern region, and so it minimizes distortion for property surveys and planning. NAD 1983 UTM Zone 18N is for regional projects in its zones so it offers good accuracy and so it was very close to the actual. US National Atlas Equal Area is used to compare areas across USA and preserves area but not designed for local accuracy and so it was off more than the previous one. NAD 1983 Contiguous USA Albers are used in environmental and demographic studies and preserves area across the USA. This was the closest to the actual area of them all. WGS 1984 World Mercator is a global projection for navigation and so it distorts area especially for from the equator. The area was off by about two times that of the actual area.

B) Map: https://drive.google.com/file/d/1yKdgVH5DKb9ZoM2RH3gi_6K92Ovvdn_C/view?usp=sharing

Multi-Frame Map of Four Global Projections



Source: Natural Earth
Credits: Shahreare Joy

C) The four map projections I used are Robinson, Goode Homolosine, Mercator, and Cube. Robinson is a good projection because it looks nice and balances shape, area, distance, and direction. The downside to this is the areas and shapes near the edges and poles are messed up and so isn't very accurate. The Goode Homolosine projection is good for showing areas accurately. The problem with this is the shapes and distances get weird and it's bad for navigation. The Mercator projection is good because it keeps the shapes and directions accurate and is good for navigation. The downside to this is that it blows up areas near the poles and so makes places like Greenland look bigger than it actually is. The cube projection breaks the globe into a cube shape and minimizes distortion on each face. This is good for displaying global data evenly without stretching or squishing areas too much. But it makes the Earth hard to see and the edges can still distort shape and distances.